

Logistic Regression

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- The statisticians approached the problem - “how can we use linear regression to solve this?”
 - We could consider the following encoding
 - Dependent variable coded as 0 (Not Spam) or 1 (Spam)

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- We can fit a linear regression to this binary response
 - and classify as **Spam** if $\hat{y} > 0.5$ and **Not Spam** otherwise, interpreting $\hat{y}$ as a probability that **Email is Spam**

on

Enforce

- maximize likelihood of



Not spam if < 0.5

A hand-drawn diagram in red ink. On the left, the letter 'x' is written. An arrow points from 'x' to a circle. Inside the circle is the letter 'y'. From the right side of the circle, three arrows point outwards to the labels '30', '1', and '2'.

$$p_{x_i} = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

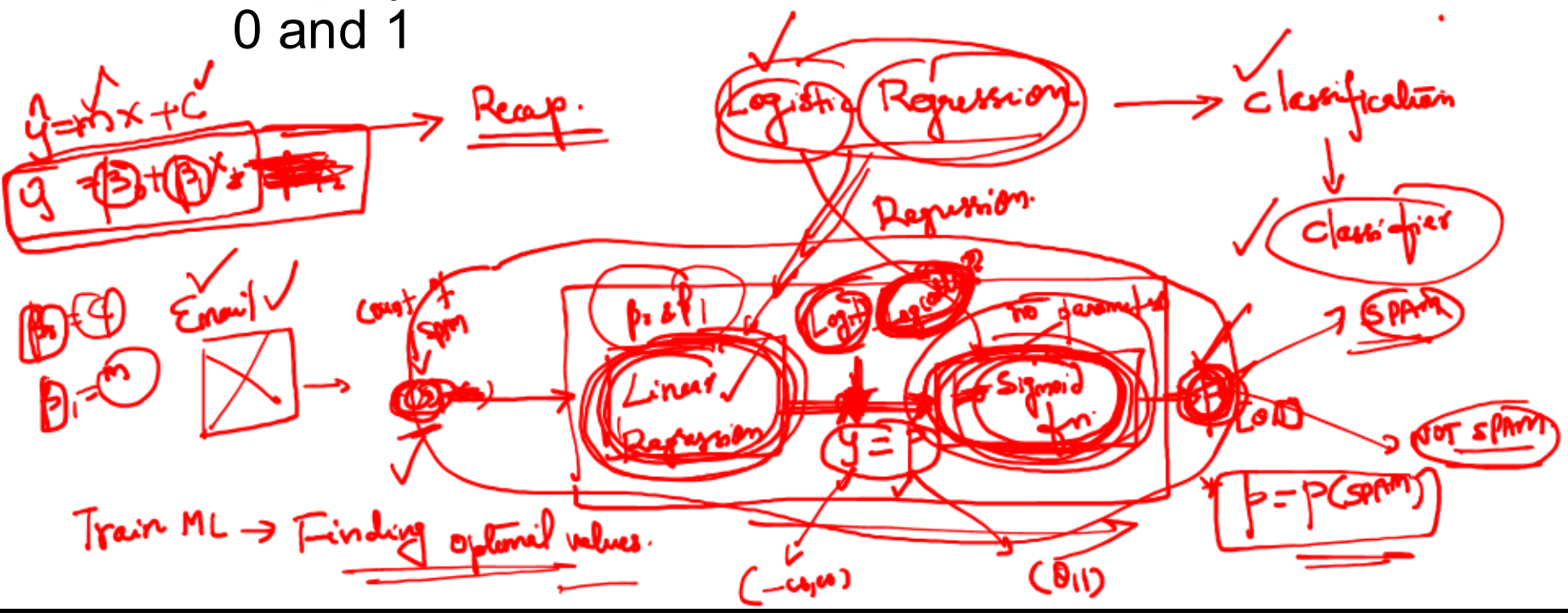
max likelihood

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- A major problem with such an approach
- Linear regression models produce values in $(-\infty, +\infty)$, which does not make sense as a probability
 - Employ a function that constrains the values between 0 and 1

MLE $\rightarrow$ maximum likelihood estimates
 $\beta_0 = ()$
 $\beta_1 = ()$



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- Loss fn / cost fn.



A graph of a sigmoid function $L(x)$ is shown. The horizontal axis is labeled x_0 and the vertical axis is labeled L . A red tangent line is drawn at the point $(x_0, L(x_0))$. The area under the curve is shaded in light blue.

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- Logistic regression model

Sigmoid output

$$p_X = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Linear regression output

$$0 \leq p_x \leq 1$$

Handwritten derivations and notes:

$e^{\log(2)} = 2$

$\log\left(\frac{p_X}{1 - p_X}\right)$ (labeled "odds")

Sigmoid Function

Probability

$$= \frac{e^{\log(p)}}{1 + e^{\log(p)}}$$

$$= \frac{\frac{p}{1-p}}{1 + \frac{p}{1-p}}$$

$$\frac{p}{(1-p) + p} = p$$

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- Terminology

- The **logit** $\log \left(\frac{p_X}{1-p_X} \right)$ ← Log odds

Handwritten example for India vs Pakistan:

	India	Pakistan
W	9	1
L	1	9

$odds(India) = \frac{0.1}{1-0.1} = \frac{0.1}{0.9} = \frac{1}{9}$
 $P(India) = \frac{9}{10} = 0.9$
 $P(Pak) = \frac{1}{10} = 0.1$

- The odds of an event is defined as

$$odds(Y = 1) = \frac{P(Y = 1)}{1 - P(Y = 1)} = \frac{p}{1 - p}$$

- Chance can be expressed either as a probability or as odds

Handwritten summary of the relationship between probability and odds:

$odds() = \frac{P()}{1 - P()}$

Probability

relative

indicate a chance for event

$\frac{9}{1} = \frac{0.9}{1-0.9} = \frac{0.9}{0.1} = \frac{9}{1}$
 $odds(India) = 9 \text{ to } 1 \Rightarrow \frac{9}{1}$
 $odds(Pak) = 1 \text{ to } 9 \Rightarrow \frac{1}{9}$

Logistic Regression

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- The cost/loss function
 - No closed form solution for MLE
 - We rely on numerical approximation to find the MLE
 - Most software uses the gradient descent algorithm ✓

Handwritten notes in red ink:

$\beta_0 =$
 $\beta_1 =$

Max

There is a red arrow pointing from the word "approximation" in the text above to the handwritten notes.



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- [illegible]

